Portfolio Builder Project

RSI Factor Analysis

Relative Strength Index and 20-Day Future Returns

Factor: RSI (14-Day)

Forward Horizon: 20 Trading Days

Portfolio Formation: Daily Cross-Sectional Deciles

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1 Objective

The objective of this analysis is to test whether the **14-day Relative Strength Index (RSI)** contains information about subsequent stock returns.

The analysis examines whether stocks with low RSI values (oversold conditions) and high RSI values (overbought conditions) have systematically different subsequent 20-day returns.

Two competing hypotheses are considered:

- **Momentum hypothesis:** stocks with high RSI should continue to outperform stocks with low RSI.
- **Reversal hypothesis:** stocks with low RSI should subsequently outperform stocks with high RSI.

The analysis uses daily cross-sectional decile portfolios, Information Coefficient (IC), long-short portfolio returns, ordinary statistical tests, and HAC/Newey-West inference.

2 Data

The analysis is based on the OHLCV stock dataset used throughout the Portfolio Builder project.

Property	**Value**
Number of stocks	24
Valid observations	34,800
Factor	14-day RSI
Forward return horizon	20 trading days
First valid RSI date	2016-09-27
Last valid observation	2022-08-05

Table 1: Dataset and analysis characteristics

For each stock and trading day, the RSI is calculated using only information available up to that day. The future return is then calculated over the following 20 trading days.

3 Research Hypothesis

The RSI is traditionally used to identify relatively strong and weak recent price movements.

For this study, the primary question is whether extreme RSI values are followed by continuation or reversal.

The reversal hypothesis can be expressed as

$$H_1 : \text{Low RSI} \rightarrow \text{Higher future returns}$$

and

$$H_1 : \text{High RSI} \rightarrow \text{Lower future returns.}$$

If this relationship exists consistently across the cross-section, the RSI factor may contain useful information for a reversal-based portfolio.

4 Factor Construction

The RSI was calculated using a 14-day lookback period.

Let the daily price change be

$$\Delta C_t = C_t - C_{t-1}.$$

The daily gain and loss are defined as

$$Gain_t = \max(\Delta C_t, 0)$$

and

$$Loss_t = \max(-\Delta C_t, 0).$$

The 14-day average gain and average loss are then calculated as

$$AvgGain_t = \frac{1}{14} \sum_{k=0}^{13} Gain_{t-k}$$

and

$$AvgLoss_t = \frac{1}{14} \sum_{k=0}^{13} Loss_{t-k}.$$

The Relative Strength is

$$RS_t = \frac{AvgGain_t}{AvgLoss_t}.$$

Finally, the RSI is

$$\boxed{RSI_t = 100 - \frac{100}{1 + RS_t}}$$

with RSI values bounded between 0 and 100.

Special cases where the average loss is zero were handled separately: if the average gain was positive, RSI was set to 100; if both average gain and average loss were zero, RSI was set to 50.

5 Future Return Construction

The dependent variable is the 20-trading-day forward return:

$$\boxed{R_{t,t+20} = \frac{C_{t+20}}{C_t} - 1}$$

where C_t is the closing price at the signal date and C_{t+20} is the closing price 20 trading days later.

The current-day RSI is therefore compared with a future return, ensuring that the factor does not directly use future price information.

6 Signal Distribution

The RSI distribution across the valid observations was:

Statistic	RSI
Count	34,800
Mean	52.7640
Standard deviation	16.8592
Minimum	1.0094
25th percentile	40.5155
Median	52.8306
75th percentile	65.1194
Maximum	99.7188

Table 2: Distribution of the 14-day RSI

The median RSI is approximately 52.83, while the mean is approximately 52.76. This indicates that the signal is broadly centered around the middle of the RSI range.

7 Decile Analysis

On every trading day, stocks were ranked according to their RSI value and divided into ten cross-sectional groups.

- G1 contains the lowest RSI stocks.
- G10 contains the highest RSI stocks.

The average 20-day future return of each group was:

Decile	Average Future 20-Day Return
G1	2.1176%
G2	1.9426%
G3	1.8471%
G4	2.0137%
G5	1.5741%
G6	1.7204%
G7	1.7201%
G8	1.3156%
G9	1.4182%
G10	0.9636%

Table 3: Average future 20-day return by RSI decile

There is a generally decreasing pattern from the lowest RSI decile toward the highest RSI decile.

The extreme-decile spread is

$$2.1176\% - 0.9636\% = \boxed{1.1539\%}.$$

Thus, the lowest-RSI portfolio produced an average future 20-day return approximately 1.15 percentage points higher than the highest-RSI portfolio.

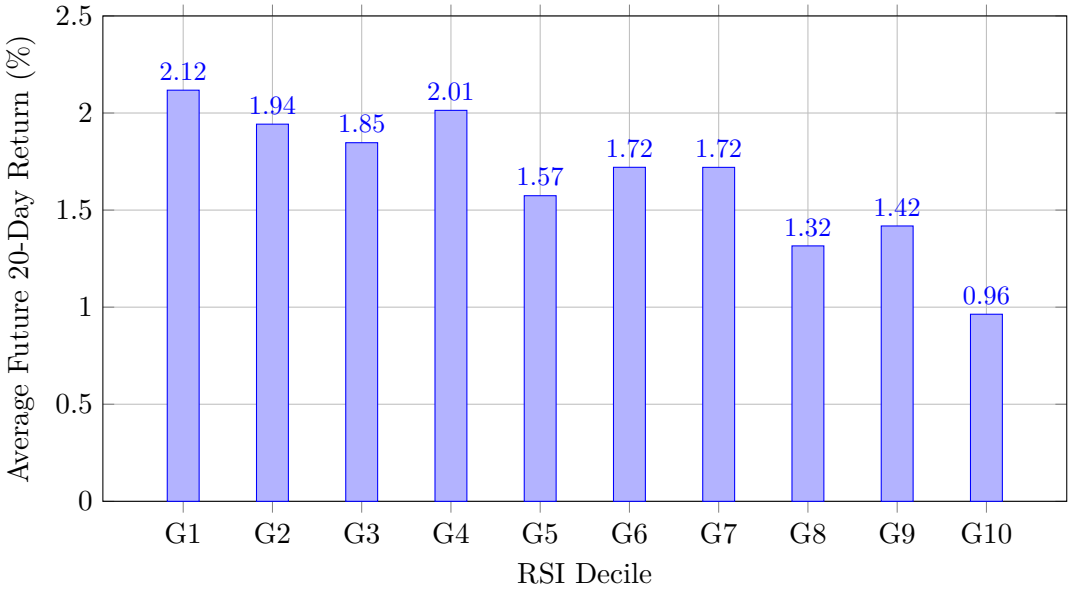


Figure 1: Average future 20-day return by RSI decile

The graph visually shows the negative relationship between RSI and subsequent returns. Lower RSI portfolios generally exhibit higher future returns than higher RSI portfolios.

8 Information Coefficient (IC) Analysis

The daily Information Coefficient measures the cross-sectional Spearman rank correlation between RSI and future 20-day returns.

The results are:

Metric	**Value**
Mean IC	-0.055168
Median IC	-0.043913
IC standard deviation	0.249441
Positive IC fraction	43.17%
t-statistic	-8.421709
p-value	< 0.001

Table 4: Information Coefficient results

The mean IC is

$$\boxed{IC = -0.055168}.$$

The negative sign indicates that higher RSI values tend to be associated with lower subsequent 20-day returns.

The positive IC fraction is only 43.17%, meaning that the daily relationship is negative more often than positive.

The ordinary t-test gives

$$t = -8.421709, \qquad p < 0.001.$$

This provides strong evidence of a negative cross-sectional relationship under the ordinary inference framework.

9 Long–Short Portfolio Analysis

A long-short portfolio was constructed as

$$LS_t = R_{G1,t} - R_{G10,t}.$$

Thus, the strategy is:

- Long the lowest-RSI decile.
- Short the highest-RSI decile.

The resulting statistics were:

Metric	**Value**
Mean daily return	0.011539
Median daily return	0.008874
Daily standard deviation	0.066563
Annualized return†	290.7913%
Annualized volatility	105.6661%
Sharpe-like ratio	2.7520
Ordinary t-statistic	6.601303
Ordinary p-value	< 0.001

Table 5: RSI long-short portfolio statistics

The average daily long-short return is

$$\boxed{0.011539}$$

or approximately

$$\boxed{1.1539\%}.$$

The positive value means that the low-RSI portfolio outperformed the high-RSI portfolio on average.

10 Overlapping Return Problem

The forward return horizon is 20 trading days, while the signal is evaluated daily.

Consequently, consecutive observations overlap.

For example,

$$R_{t,t+20}$$

and

$$R_{t+1,t+21}$$

share 19 trading days of their return periods.

This creates serial correlation in the daily long-short return series.

As a result, ordinary standard errors can underestimate the true uncertainty of the estimated mean return.

Therefore, ordinary t-statistics should not be considered sufficient for final inference.

11 HAC / Newey–West Test

To account for serial correlation caused by overlapping 20-day returns, a HAC/Newey-West standard error with 20 lags was used.

The results were:

Metric	HAC Result
Mean return	0.011539
HAC standard error	0.004311
HAC t-statistic	2.676568
HAC p-value	0.007438

Table 6: HAC/Newey-West inference

The HAC test gives

$$t_{HAC} = 2.676568$$

and

$$p_{HAC} = 0.007438.$$

The p-value remains below 0.05 even after accounting for the overlapping-return problem.

This indicates that the negative RSI–future-return relationship remains statistically significant under HAC-adjusted inference.

12 Comparison of Statistical Tests

The main statistical results can be summarized as follows:

Test	Statistic	p-value
IC ordinary t-test	-8.421709	< 0.001
Long-short ordinary t-test	6.601303	< 0.001
Long-short HAC test	2.676568	0.007438

Table 7: Comparison of RSI statistical tests

The ordinary tests indicate a strong relationship. More importantly, the HAC-adjusted long-short test remains statistically significant.

Therefore, the result is not solely a consequence of ignoring the serial correlation generated by overlapping returns.

13 Economic Interpretation

The empirical evidence supports a **reversal interpretation** of the RSI signal in this dataset.

The key observation is:

$$\text{Low RSI} \quad \longrightarrow \quad \text{Higher subsequent 20-day return}$$

while

$$\text{High RSI} \quad \longrightarrow \quad \text{Lower subsequent 20-day return.}$$

The lowest RSI decile generated a future return of 2.1176%, whereas the highest RSI decile generated only 0.9636%.

The resulting extreme-decile spread was 1.1539 percentage points.

The negative mean IC of -0.055168 provides an additional indication that the relationship is systematically inverse rather than a single isolated decile effect.

The HAC p-value of 0.007438 further indicates that the relationship remains statistically significant after accounting for overlapping returns.

14 Interpretation of the Annualized Return

The mechanically annualized long-short return is reported as 290.7913%.

However, this number should **not** be interpreted as a realistic expected annual return.

The reason is that the underlying 20-day forward returns overlap when observations are calculated every trading day.

The annualized figure is therefore included as a mechanical statistic for comparison with other factor tests, rather than as a forecast of actual annual portfolio performance.

The Sharpe-like ratio should also be interpreted cautiously for the same reason.

15 Limitations

Several limitations should be considered before using RSI as a portfolio signal.

15.1 Overlapping Returns

The 20-day forward return creates overlapping observations. HAC adjustment addresses the statistical inference problem but does not remove the economic dependence between observations.

15.2 Single Stock Universe

The analysis is based on the available 24-stock universe. Therefore, the results may not generalize to a broader market universe.

15.3 Transaction Costs

The analysis does not include brokerage costs, bid-ask spreads, slippage, taxes, or market impact.

A practical strategy would need to account for these costs.

15.4 Signal Redundancy

RSI is derived from price movements and may contain information similar to other reversal-related signals such as momentum, moving-average deviation, breakout measures, and trend strength.

Therefore, the RSI signal should eventually be tested jointly with other factors.

15.5 Out-of-Sample Validation

The current analysis is an in-sample historical factor test. An out-of-sample or walk-forward evaluation is required before drawing conclusions about future performance.

16 Factor Classification

Based on the current tests, RSI is classified as a:

Primary Reversal Factor

The classification is supported by three observations:

1. The RSI decile returns show a generally decreasing pattern.
2. The mean IC is significantly negative.
3. The HAC-adjusted long-short return remains statistically significant.

The factor therefore provides stronger evidence than signals for which the relationship disappears after HAC adjustment.

17 Conclusion

The 14-day RSI factor exhibits a strong negative relationship with subsequent 20-day stock returns in the tested dataset.

The lowest RSI decile generated an average future return of

$$2.1176\%$$

while the highest RSI decile generated

$$0.9636\%.$$

The resulting long-short spread was

$$\boxed{1.1539\%}.$$

The mean Information Coefficient was

$$\boxed{-0.055168},$$

and the ordinary IC test produced

$$p < 0.001.$$

Most importantly, the HAC/Newey-West test produced

$$t = 2.676568, \qquad p = 0.007438,$$

which remains statistically significant after accounting for overlapping 20-day returns.

Therefore, within this dataset and methodology, RSI provides evidence of a **reversal effect**: stocks with relatively low RSI tend to have higher subsequent 20-day returns than stocks with relatively high RSI.

For the Portfolio Builder project, RSI is therefore retained as a **primary reversal factor** and can be considered in the next stage of multi-factor portfolio construction.